

Attaining Consumer Insights with Snowflake and Microsoft Power BI – Part 2

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MSDS-535: Data Mining

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Date: Summer 2026

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Introduction

Retail and consumer packaged goods (CPG) companies rely on data to understand customer behavior, product performance, sales channels and profitability. Point-of-sale (POS) data is one of the richest sources of business intelligence in retail environments. It gives visibility into what products are selling, through which channels, and at what margin. But reliable and timely sales data can be elusive as retailers must often rely on distributors, third-party providers and outside data sources to fill the gaps.

The lab “Attaining Consumer Insights with Snowflake and Microsoft Power BI” showed how Snowflake can solve this problem by integrating internal enterprise data with live third-party marketplace data in a single cloud platform. Snowflake, together with Microsoft Power BI, offers a robust environment to create dashboards, perform analytics and empower business users with self-service reporting.

In this report, we summarize how organizations can access first-party data in Snowflake, leverage the Snowflake Data Marketplace for live POS data, and develop consumer insights with Power BI using self-service analytics.

Accessing First-Party Data Seamlessly in Snowflake

First-party data is data that an organization collects directly from its operations, systems, or customer interactions. In retail and CPG environments this can include:

- sales deals
- customer buying history
- product stock
- pricing information promotions information internal financial reporting
warehouse or distribution records

Snowflake offers a way for organizations to unify this data in a single cloud-native platform with ingestion pipelines, staging layers, and warehouse storage. Snowflake's decoupled storage and compute allow teams to efficiently query large datasets and scale across multiple workloads.

One of the major advantages we could demonstrate in the lab was the ability to seamlessly access internal datasets without the need to move data across multiple tools or siloed systems. Business users, data analysts and data teams can all work off of the same source of truth, leveraging Snowflake's SQL engine and cloud storage architecture.

Han et al. (2022) reported that modern analytics depend heavily on centralized data warehouse environments, which combine data from multiple sources into a common structure that is optimized for decision-making and reporting.

Leveraging the Data Marketplace to Query Live POS Data

One of the main advantages of Snowflake is the Snowflake Data Marketplace which gives organizations the ability to access third-party data without the need to manually acquire, transfer or manage the files themselves.

This was demonstrated in the lab using live point-of-sale data from external providers. This data can be:

- retailer sales performance market trends
- geographic sales activity product movement by channel consumer demand trends
- competitor benchmarking data

This type of information is usually conveyed through individual vendor contracts, manual file exchanges, or delayed reporting processes. Snowflake makes it easier for organizations to subscribe to marketplace datasets and query them directly in conjunction with their internal enterprise data.

This offers powerful analytical opportunities as companies can combine:

Internal Data

- Sales promotions, internal Margin performance Stock

Marketplace Data External

- POS trends channel performance regional sales intelligence market benchmarks

You can see more of what your customers are doing and how your business is performing when you can query both side by side.

Turning Data into Consumer Insights with Power BI

Once the data is in Snowflake, you can connect Microsoft Power BI directly to your Snowflake warehouses to create dashboards and visual analytics.

Power BI offers interactive reporting features like:

- charts dashboards
- KPI tracking filters Drill down analysis
- reporting self-service

With this integration, business users can get from raw data to actionable insight fast.

Power BI enables retail and CPG organizations to answer questions such as:

- What products are selling the quickest?
- Which retail channels are driving the most revenue?
- Where is the demand the greatest?
- Which products have the greatest margin?
- "How are promotions affecting sales?
- Where are inventory shortages happening?

This powerful combination of Snowflake's scalable analytics engine with Power BI's visualization capabilities can be used by organizations to build near real-time decision support systems for leadership teams.

Enabling Self-Service Analytics

One of the most useful results of the lab was the support for self-service analytics.

Self-service analytics enables business users to work with and analyze data without requiring technical teams to manually craft each report.

This means that users across the business, including operations, finance, sales, marketing and leadership can:

- view dashboards apply filters
- See the trends
- answer questions independently create reports

These cuts reporting bottlenecks and improves decision-making speed.”

Instead of waiting for technical teams to generate reports, users can now interact directly with dashboards and trusted datasets in Power BI connected to Snowflake.

Aggarwal (2015) states that the value of a business intelligence system is maximized in organizations that can rapidly translate raw enterprise data into intelligible insights that can assist in strategic decision-making. Self-service analytics is an enabler straight up.

Business Benefits of Snowflake and Power BI Integration

The Snowflake and Power BI integration demonstrated several clear advantages for retail and consumer-focused organizations.

Centralized Data Access

All relevant datasets can be accessed from one cloud platform.

Real-Time Insights

Organizations can query fresh data rather than relying on delayed reporting cycles.

Improved Decision-Making

Executives can make faster and more informed business decisions using current data.

Easier Third-Party Data Access

Snowflake Data Marketplace simplifies access to external retail and POS datasets.

Scalable Analytics

Snowflake supports enterprise-scale performance across large datasets.

Better Visualization

Power BI turns complex data into intuitive dashboards for non-technical users.

Together, these benefits make Snowflake and Power BI a highly effective combination for consumer analytics and retail intelligence.

Conclusion

The Attaining Consumer Insights with Snowflake and Microsoft Power BI lab showed how organizations could mix cloud-native data warehousing with business intelligence tools to create actionable consumer insights.

Snowflake helps enterprises consolidate internal enterprise data, as well as access live third-party POS data through the Data Marketplace. With Microsoft Power BI, these datasets can be converted into dashboards, reports, and self-service analytical tools for business users.

This combination offers more insight into product performance, channel activity, customer behavior and market trends for retail and CPG organizations. It also democratizes analytics across the business and reduces reliance on fragmented reporting processes.

In summary, Snowflake and Power BI together offer a scalable and modern solution for transforming enterprise and consumer data into valuable business intelligence.

References

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